Project Part 3 – Statistical Analysis Plan

Project so far

- `MED_EARNINGS`: median earnings for graduates from each university
- `CONTROL`: whether the school is public or private
- `SATVRMID`: midpoint of SAT critical reading scores of students attending the school
- `SATMTMID`: midpoint of SAT math scores
- `SATWRMID`: midpoint of SAT writing scores
- `ACTCMMID`: midpoint of the ACT cumulative scores
- `ACTENMID`: midpoint of ACT English scores
- `ACTMTMID`: midpoint of ACT math scores
- + others...

> Research goal: want to investigate relationship between different school characteristics and graduate earnings.

Project Parts 3 and 4

Suppose a high school senior is looking through the college scorecard data, and they have some questions for you about the value of different schools.

- Do students who graduate from more expensive schools earn more money after graduation, after accounting for student debt, school type (public or private), and the test scores of enrolled students?
- Is the relationship between school cost and graduate earnings the same for public and private schools, after accounting for student debt and the test scores of enrolled students?

Part 3: Create a Statistical Analysis Plan, describing how you will analyze the data to address these questions. Due: April 15.

Part 4: Carry out your SAP, and present your results in a full written report. Due: April 29 (tentative).

Statistical Analysis Plan (SAP)

A **statistical analysis plan** (SAP) describes:

- the research questions we are trying to answer
- the data we will use to answer the research questions
- how we will use statistical methods (models, diagnostics, hypothesis tests, etc.)
- how we will present our results

Why?

- having a plan ahead of time helps organize our work
- research questions should inform analysis and methods (not the other way around)
- helps determine which data we need to collect
- make sure we're doing something reasonable before we do a ton of work!

Statistical Analysis Plan (SAP)

- What is/are the research question(s) or objectives?
- What are the variables that will be measured in the study? For each variable, what will the data look like?
- What is the study design, and why is the study set up this way?
- How will data and any patterns/results be summarized? (i.e., how you will do exploratory data analysis).
- What will be examined statistically? Link directly back to the research questions.
- What alternative strategies should be considered, and when?
- What are the statistical results that will be presented?

Today

Today, we will focus on a few of these questions in class:

- What are the variables that will be measured in the study?
- How will data and any patterns/results be summarized? (i.e., how you will do exploratory data analysis).
- What will be examined statistically?
  - Which models will you fit?
  - How will you assess the suitability of these models? (diagnostics, etc.)
  - How will you use the models to address the research question(s)? (hypothesis tests, confidence intervals, etc.)
  - What software will you use?
- What alternative strategies should be considered, and when? (i.e., what will you do if assumptions are violated? How would you modify the model or statistical analysis?)

Variables

- Do students who graduate from more expensive schools earn more money after graduation, after accounting for student debt, school type (public or private), and the test scores of enrolled students?
- Is the relationship between school cost and graduate earnings the same for public and private schools, after accounting for student debt and the test scores of enrolled students?

> What variables should we consider to investigate these research questions?

salary 10 years after graduation,
cost to attend school,
public vs. private,
debt,
test scores (look at some combo or cumulative test score variable)

EDA

- Do students who graduate from more expensive schools earn more money after graduation, after accounting for student debt, school type (public or private), and the test scores of enrolled students?
- Is the relationship between school cost and graduate earnings the same for public and private schools, after accounting for student debt and the test scores of enrolled students?

> What EDA would you do?

- check missingness, # of different observations per level of categorical variable
- look at univariate distributions for quantitative variables (e.g. histogram)
- look at scatterplots between quantitative variables

Statistical methods

- Do students who graduate from more expensive schools earn more money after graduation, after accounting for student debt, school type (public or private), and the test scores of enrolled students?
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> Which models will you fit?

response: earnings
explanatory: all the other variables discussed

response: earnings
explanatory: all the other variables discussed, <u>and</u> an interaction between cost and public vs. private

Statistical methods

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- Is the relationship between school cost and graduate earnings the same for public and private schools, after accounting for student debt and the test scores of enrolled students?

> How will you assess the suitability of these models? (diagnostics, etc.)

- residual plot to check shape & constant variance
- QQ plot of residuals to check normality
- VIFs to check for multicollinearity
- studentized residuals to check for outliers

Statistical methods

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> How will you use the models to address the research question(s)? (hypothesis tests, confidence intervals, etc.)

Statistical methods

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- Is the relationship between school cost and graduate earnings the same for public and private schools, after accounting for student debt and the test scores of enrolled students?

> What alternative strategies should be considered, and when? (i.e., what will you do if assumptions are violated? How would you modify the model or statistical analysis?)